Questions by topic: Principles of pollution control (CPA, CGM, new technologies) - Set 1 (75 marks)

Answer all questions in the spaces provided.

0 1 A nuclear reprocessing plant on the coast discharges low-level liquid radioactive waste into the sea through a pipeline.

0 1 . 1 Explain what is meant by **Critical Pathway Analysis** (CPA).

[2 marks]

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0 1 . 2 Give **three** factors that would be considered in a CPA for this discharge.

[3 marks]

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0 1 . 3 Describe how **Critical Group Monitoring** (CGM) would be used to assess the risk to the public from this discharge.

[3 marks]

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0 1 . 4 Suggest **two** types of environmental sample that would be collected to check the CPA predictions.

[2 marks]

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0 2 . 1 Explain how the control of emission **location** and emission **timing** can reduce the severity of pollution. Use examples.

[6 marks]

Emission location .....

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Emission timing .....

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0 2 . 2 Explain the **polluter pays** principle and suggest why it encourages pollution prevention.

[2 marks]

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0 2 . 3 Explain the **precautionary principle** as applied to the release of a new chemical.

[2 marks]

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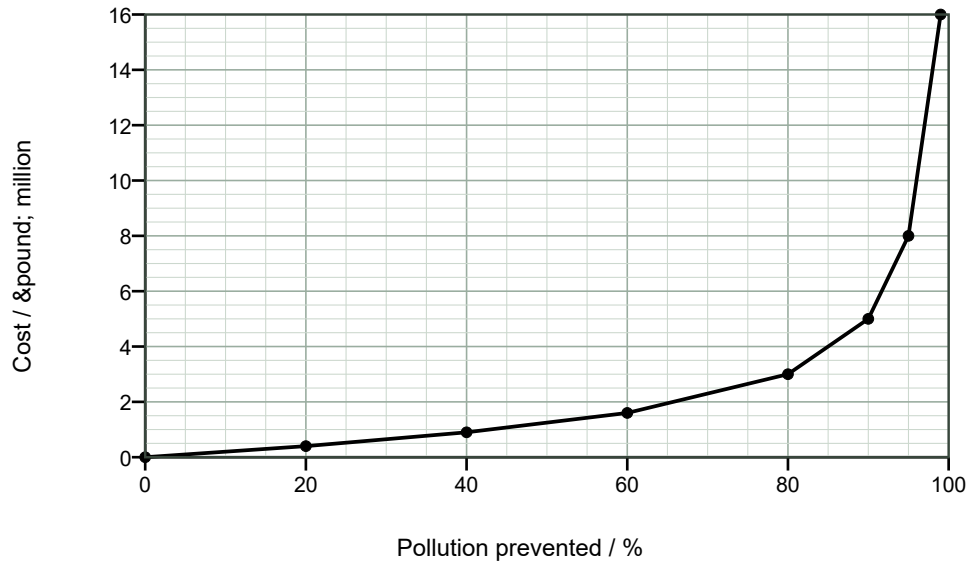
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0 3 Figure 1 shows the relationship between the cost of pollution control equipment at a factory and the percentage of a pollutant emission that is prevented.

Figure 1



0 3 . 1 Describe the relationship shown in Figure 1.

[2 marks]

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0 3 . 2 Use Figure 1 to calculate the extra cost of increasing the pollution prevented from 90% to 99%.

[1 mark]

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Answer [ ] £ million

0 3 . 3 Explain what is meant by the terms **ALARA** and **BATNEEC** and how they would be applied by the factory.

[4 marks]

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0 3 . 4 Suggest why the control methods used originally by a factory may no longer be sufficient after several years.

[3 marks]

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0 4 Table 1 shows four general approaches to pollution control.

0 4 . 1 Complete Table 1 by giving one example of each approach.

[4 marks]

Table 1

Approach	Example
Production prevention	
Prevention of release	
Post-release remediation	
Alternative processes	

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0 4 . 2 Explain why prevention of production is generally preferable to post-release remediation.

[2 marks]

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0 4 . 3 Describe how each of the following new technologies helps to control pollution.

[4 marks]

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Satellite monitoring of oil spills .....

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GPS ship tracking .....

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Polymer adsorption of heavy metals .....

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Phytoremediation .....

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0 5 . 1 Which of the following best describes the purpose of Critical Group Monitoring? Tick one box.

[1 mark]

A To detect health damage in the population most affected by a pollutant ☒

B To predict the route a pollutant will take through the environment ☐

C To assess exposure of the people most at risk so that everyone else can be assumed to be sa ☒

D To measure the concentration of a pollutant at the point of release ☐

0 5 . 2 Explain why CPA can reduce the cost of an environmental monitoring programme.

[2 marks]

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0 5 . 3 A new chemical has properties similar to PCBs (persistent, liposoluble chlorinated organic compounds). Explain how knowledge of PCBs helps decisions about the new chemical.

[2 marks]

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0 6 . 1 Which of the following best describes the **precautionary principle**? Tick **one** box.

[1 mark]

A A waste is assumed to cause pollution until research shows it is unlikely to do so ☒

B The polluter must pay for any damage caused ☐

C Emissions should be as low as reasonably achievable ☐

D Only pollutants known to be toxic need to be controlled ☐

0 6 . 2 Explain why the polluter pays principle works best when penalties are high and enforcement is reliable.

[2 marks]

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0 6 . 3 Give **two** choices about emission location that reduce the severity of pollution.

[2 marks]

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0 7 A nuclear fuel reprocessing plant discharges small amounts of radioactive material in liquid effluent to a coastal sea.

0 7 . 1 Explain why it cannot be assumed that the discharge will be diluted to insignificant levels.

[2 marks]

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0 7 . 2 Give **four** factors that would be included in a Critical Pathway Analysis of this discharge.

[4 marks]

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0 7 . 3 Explain how the results of the analysis reduce the cost of environmental monitoring.

[2 marks]

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0 7 . 4 Explain why Critical Group Monitoring measures exposure rather than damage to health.

[2 marks]

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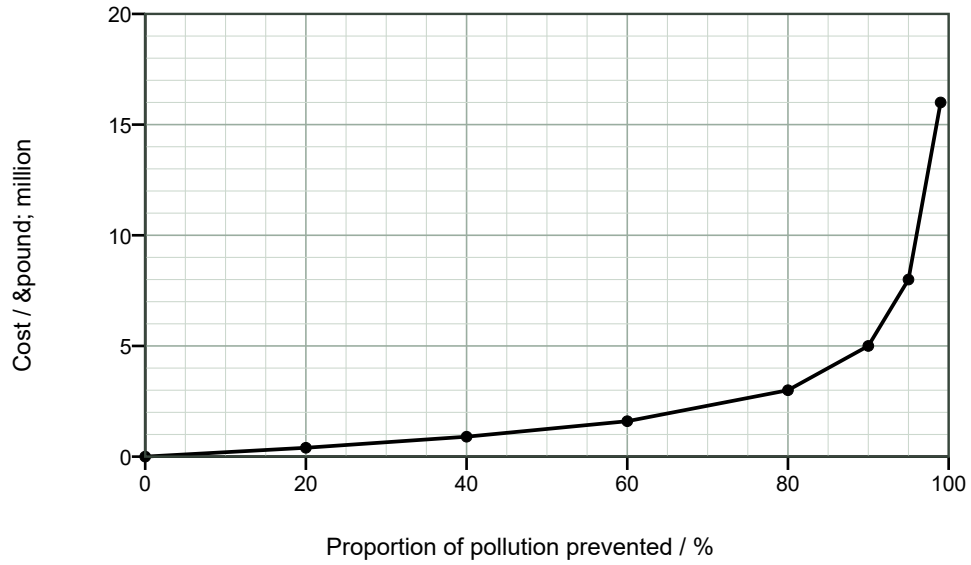
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0 8 Figure 1 shows how the cost of controlling emissions from a factory changes with the proportion of the pollutant that is prevented from being released.

Figure 1



0 8 . 1 Use Figure 1 to calculate the extra cost of increasing control from 90% to 99% of the pollutant.

[1 mark]

Answer £ million

0 8 . 2 Explain how the terms **ALARA** and **BATNEEC** are used when deciding how much pollution control is required.

[3 marks]

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0 8 . 3 Explain why the control methods that were originally adequate for a factory may no longer be sufficient.

[2 marks]

0 8 . 4 Evaluate the principles and general strategies that can be used to decide how pollution should be controlled.

[9 marks]

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Extra space

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END OF QUESTIONS